Problem 00000001470: no finite exact level

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Result and scope. For the fixed input polynomials $g_1 = X^3$ and $g_2 = 1 - X^2$, the compact basic closed set is $K = [0, 1]$. The prescribed Lasserre moment projection strictly contains K at every legal level $r \geq 2$. For the objective X, both the actual moment infimum and the truncated SOS supremum are strictly below the true minimum 0 at every such level. Thus there is no finite exact level for this input, and no universal finite-convergence bound of the printed form.

This concerns the hierarchy built from the given defining tuple. It does not assert that $[0, 1]$ lacks some other finite SDP representation: it plainly has one. Changing the generators, adding optimality equations, or imposing additional generic regularity conditions gives a different assertion. We interpret the source's explicit parenthetical "universal bound for finite convergence" in this standard input-dependent hierarchy sense.

1. Actual matrices and polynomial tests

We use real linear functionals $L : \mathbb{R}[X] \to \mathbb{R}$, with actual monomial moments $y_j = L(X^j)$. At level $r \geq 2$, feasibility is $L(1) = 1$ and positive semidefiniteness of the three finite real matrices

$$M_r(L) = (y_{i+j})_{0 \leq i,j \leq r}, \quad M_{r-2}(X^3 L) = (y_{i+j+3})_{0 \leq i,j \leq r-2}, \quad M_{r-1}((1 - X^2)L) = (y_{i+j} - y_{i+j+2})_{0 \leq i,j \leq r-1}.$$

Only moments through degree $2r$ occur. These are the usual moment and localizing matrices with the ceiling-half-degree offsets; see Nie [1], printed pp. 1–2, (1.2)–(1.4). The proof constructs genuine full linear functionals, so in particular supplies the required finite moment vectors.

Lean defines each matrix entry as $L(gX^{i+j})$, proves it Hermitian, and proves its quadratic form is

$$a^T M_k(gL) a = L\left(g \left(\sum_{i=0}^{k} a_i X^i \right)^2 \right).$$

The converse uses the actual coefficient expansion of every degree-k polynomial. Consequently the matrix is `Matrix.PosSemidef` if and only if $L(gp^2) \geq 0$ for every $\deg p \leq k$. The matrices are not numerical labels or assumed positivity predicates.

2. A negative feasible moment at every level

Fix $d = r - 2$ and the $d + 1$ distinct nodes $t_i = (i + 1)/(d + 2) \in (0, 1)$. Lagrange interpolation, Cauchy–Schwarz and compactness of $[-1, 0]$ give a constant $C > 0$ such that

$$p(t)^2 \leq C \sum_i p(t_i)^2 \qquad (t \in [-1, 0],\ \deg p \leq d).$$

Set $w_i = C/t_i^3 > 0$, $A = \sum_i w_i t_i \geq 0$, $\varepsilon = 1/(A + 1) \in (0, 1]$, and $m = 1 + \varepsilon^3 \sum_i w_i > 0$. Define the genuine positive atomic functional

$$L(p) = \frac{p(-\varepsilon) + \varepsilon^3 \sum_i w_i p(t_i)}{m}.$$

It satisfies $L(1) = 1$ and is nonnegative on every square. All atoms lie in $[-1, 1]$, hence $L((1 - X^2)p^2) \geq 0$ for every polynomial. For degree $\leq d$, interpolation gives

$$mL(X^3 p^2) = \varepsilon^3 \left(C \sum_i p(t_i)^2 - p(-\varepsilon)^2 \right) \geq 0.$$

Thus all three actual matrices are PSD. However,

$$mL(X) = -\varepsilon + \varepsilon^3 A < 0, \qquad \varepsilon^2 A = \frac{A}{(A + 1)^2} < 1.$$

Every finite level therefore has a feasible negative first moment. Evaluation at any actual $x \in K$ is also proved feasible. It follows that the first-moment projection strictly contains K, not merely that a chosen certificate search fails.

3. Genuine optimization values and an SOS gap

Let $V_r = \{L(X) : L \text{ is moment feasible}\}$. Let D_r be the real γ for which actual finite sums of polynomial squares satisfy

$$X - \gamma = \sigma_0 + X^3\sigma_1 + (1 - X^2)\sigma_2,$$

where the squared polynomials have degree at most $r, r-2, r-1$, respectively. Lean's sums-of-squares definition includes the actual polynomials and finite sum identities.

The identity

$$X + 1 = \tfrac{1}{2}(X + 1)^2 + \tfrac{1}{2}(1 - X^2)$$

proves $-1 \in D_r$ for every $r \geq 2$. Each half-square is encoded as two squares of a polynomial with coefficient $1/2$. Applying any feasible L to any such SOS identity gives the proved weak-duality inequality $\gamma \leq L(X)$. In particular V_r is nonempty and bounded below by -1, and D_r is nonempty and bounded above. Choosing the negative feasible functional from Section 2 gives, for all $\gamma \in D_r$, the same strict negative upper bound $L(X) < 0$.

Lean defines the real values by actual $\inf V_r$ and $\sup D_r$ and proves

$$-1 \leq \inf V_r < 0, \qquad -1 \leq \sup D_r < 0.$$

The nonempty and boundedness proofs exclude empty-set or unbounded-set fallback values. No optimizer attainment, strong duality, or mere nonexistence of a certificate is used to deduce the strict gaps. Separately, the actual semialgebraic set is proved compact and its actual objective image has least element 0. Hence neither hierarchy value is exact at any legal finite level. A fixed degree-three input failing at every level refutes any universal finite bound depending on that degree.

4. Verification and attribution

All 58 named declarations (43 theorems and the remaining definitions) are included in the axiom audit. The package contains complete original bilingual source, this LaTeX/PDF, a pinned Lean project, fresh-build and official fresh-kernel replay logs, and SHA256 manifests. No `sorry`, `native_decide` or additional axiom is used. Separate AI supervision reviews the source semantics and all rendered PDF pages. Organizer acceptance and human verification are not asserted.

The interpolation/atomic construction and all finite-level conclusions are proved directly; the primary reference supplies the standard hierarchy definitions, not an imported nonconvergence axiom. No earlier same-ID submission was found in the recorded search. This selective production result is not an additional sample in the already frozen workflow or CPU experiments.

References

[1] J. Nie, *Certifying Convergence of Lasserre's Hierarchy via Flat Truncation*, Mathematical Programming 142 (2013), 485–510, author PDF, printed pp. 1–2, equations (1.2)–(1.4), for the moment/SOS hierarchy and degree offsets.